Highlights

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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- Dual-mode SUVs track surface winds and sample subsurface currents.
- Event-specific ERA5–HYCOM environments cover a 500-TC archive.
- Parameter-sharing IPPO scales TC tracking across 4–16 SUVs.
- Active underwater control increases current exposure and sampling footprint.

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

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ABSTRACT

Tropical cyclones (TCs) are destructive, while intensity and rapid-intensification forecasts remain limited by sparse near-core observations. Platforms rarely combine rapid transit, surface-wind sampling, and subsurface-current sensing. We develop simulation-based parameter-sharing independent proximal policy optimization (IPPO) for coordinated observation with Blue Whale-class submersible unmanned vessels. Event-specific environments align ERA5 10 m winds, HYCOM currents from 0 to 60 m, and IBTrACS tracks for a 500-TC western North Pacific archive. The surface policy uses shared winds; the underwater policy uses each vehicle's local current history. Decisions occur every 10 min, and the TC center enters only the training reward. Final tracking error, the end-of-episode vessel-to-center distance on unseen TCs, is an operational proxy for observation value. The 20 m/s policy lowers final tracking error relative to 12 m/s in all 20 fleet-size/radius comparisons; 12 m/s yields shorter, lower-speed paths. Low-pass filtering further lowers fleet-mean final error, path length, and peak speed. An interim seven-depth test on 23 common held-out TCs compares active IPPO, random control, and passive drift. Active IPPO raises mean experienced current speed by 1.6–5.6% and path length by 3.7–7.8 times versus passive drift. Active 2.0 m/s paths are 1.24–1.32 times longer than 1.5 m/s paths. Results support active surface tracking and subsurface sampling.

1. Introduction

TCs threaten coastal communities, infrastructure, and offshore operations, and the proportion of intense and rapidly intensifying events has increased in many basins [1, 2, 3, 4]. Landfalling TCs over the western North Pacific and China have also become more destructive [5, 6, 7, 8]. Track prediction has advanced through numerical weather prediction, satellite assimilation, ensembles, and data-driven models, whereas intensity prediction remains more difficult [9, 10]. A central reason is that intensity depends on air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier layers, mesoscale eddies, and TC-induced currents [11, 12]. Yet clouds and rain limit microwave sensing, scatterometers saturate at high winds, and spaceborne sensors cannot resolve the subsurface ocean. Near-core wind and upper-ocean states therefore remain poorly constrained in many forecast initial conditions [13, 14, 15].

Existing observing platforms each cover only part of this problem. Satellites provide broad coverage but cannot simultaneously observe the surface wind maximum and three-dimensional upper ocean [13]; aircraft obtain high-quality near-core measurements but are expensive and campaign-limited [14, 15]; and gliders can remain at sea for long periods but move much more slowly than a translating TC [16]. Autonomous underwater vehicles (AUVs) can adapt their sampling to changing environmental and vehicle states [17]. Upper-ocean observations can improve initial states and, in some cases, intensity forecasts [18, 19, 20, 21, 22], but their value is highly sensitive to sampling location and background structure [23, 24, 25]. Adaptive TC observation is therefore a sequential decision problem involving motion, sampling time, energy, and coordination rather than coverage alone.

A submersible unmanned vessel (SUV) combines rapid surface transit with subsurface sensing [26, 27, 28, 29]. The Blue Whale is used as an engineering reference: its public specifications indicate a surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s), and a reference depth near 60 m [26]. Rather than claiming a digital twin, this study parameterizes identifiable future Blue Whale-class capabilities. In the proposed mission, the surface fleet observes 10 m winds, follows the TC toward the high-wind region as closely as possible, and is evaluated by its final center distance. At depth, four SUVs are assigned to each prescribed layer and seek the strongest observable current. Figure 1 illustrates this handoff; four vehicles are shown at the surface as an example, and the same parameter-sharing policy is later evaluated with 4, 8, 12, and 16 SUVs.

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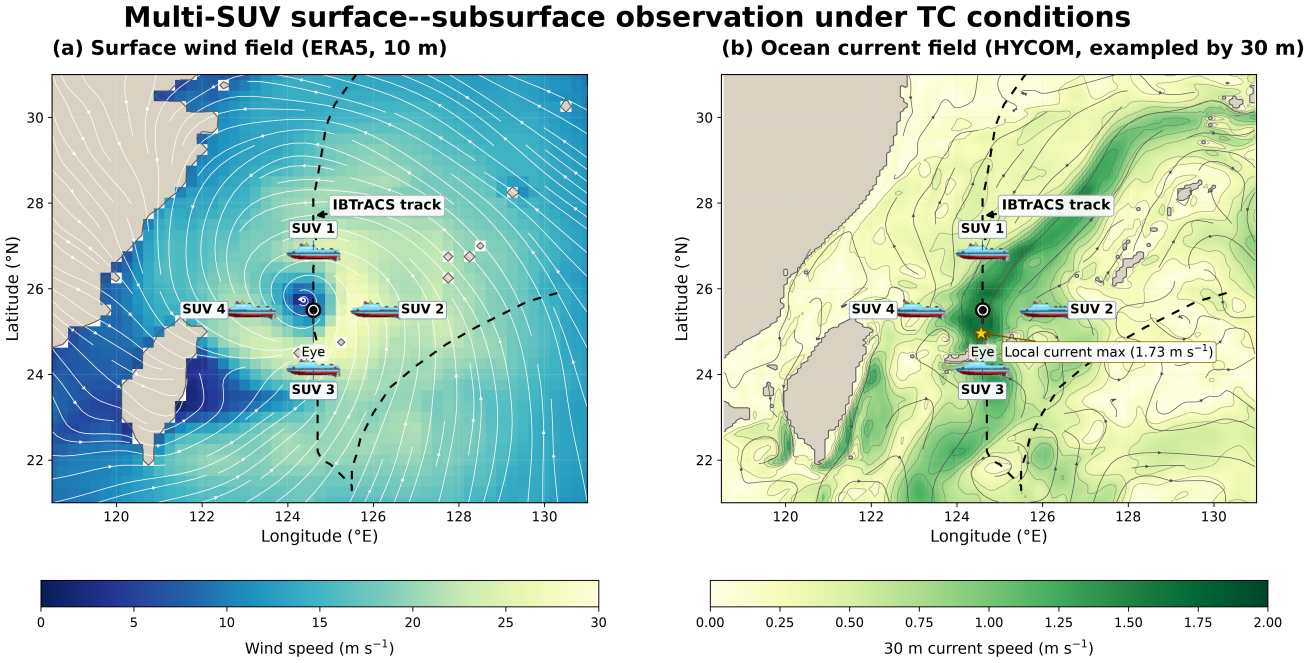


Figure 1: Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

Classical graph search, sampling-based planning, potential fields, and model predictive control are effective when the cost map and dynamics are known [30, 31, 32, 33, 34, 35]. In a TC, however, the fields are nonstationary and only locally observed, while propulsion is superimposed on ambient flow. Conventional pipelines also estimate a field or waypoint sequence before execution. By contrast, reinforcement learning (RL) maps each new measurement directly to the next action, closing the observation–decision–motion loop. RL has demonstrated strong sequential decision making across diverse domains [36, 37, 38, 39, 40, 41, 42, 43], and independent PPO is competitive in cooperative multi-agent tasks [44, 45, 46]. Multi-AUV adaptive sampling and target tracking provide the ocean-robotics foundations [47, 48, 49, 50, 51].

Three gaps remain. First, prior TC observation studies often represent the ocean with analytical or reduced-order flows rather than event-specific fields [52, 53]. Second, track-forecast errors require rapid, adaptive transit from an uncertain deployment point to the high-wind region [23, 24, 25]. Third, the closest multi-AUV active-observation study used only four vehicles [53]. We therefore formulate stochastic policy optimization under partial observability [54, 44, 45], extend the earlier center-inference ideas [52, 53] to event-specific surface–subsurface observation, and evaluate 4-, 8-, 12-, and 16-SUV fleets.

The main contributions of this study are summarized as follows.

1. We construct an event-specific learning environment for 500 TCs by aligning ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks.
2. Wind-only parameter-sharing IPPO learns inward motion from training-time center rewards and lowers the final tracking error, measuring how closely the surface fleet follows the TC and enters the high-wind region. The actor never receives the center coordinate; during validation it infers an implicit target direction from shared wind observations.
3. We compare active IPPO, random control, and passive drift for subsurface observation with four SUVs at each depth, using the highest attainable observed current speed as the objective.
4. We quantify how experienced current speed and sampling displacement vary over seven depths from 0 to 60 m, separating vertical attenuation of the TC response from the added mobility produced by active control.

2. Problem Formulation

2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. At the surface, each SUV uses its environmental observation to move toward the high-wind region around the moving

TC; the best-track center is used for training rewards and evaluation, not as an actor input. The subsurface stage is then initialized within the 200 km near-core region, where four SUVs are assigned to a prescribed depth and seek the highest attainable current speed. The framework is designed to satisfy four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. cooperative tracking of the moving TC through decentralized action selection, using onboard wind observations and idealized inter-vehicle sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe and controllable operation under the specified energy, communication, smoothing, and coastline settings.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is 1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: surface propulsion is used to follow the TC and reach the high-wind region as closely as possible, quantified primarily by the final tracking error. At depth, the objective changes to finding and sampling the highest possible current speed, so the four SUVs must weigh active propulsion against passive advection by the ambient current.

2.2. Problem Definition

The fleet contains N SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \ldots, N\}$. The position, propulsion velocity, and operating mode of SUV i at time t are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. The propulsion speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V_{\max}^{\mathrm{s}}, & m_i(t) = \mathrm{s}, \\ V_{\max}^{\mathrm{u}}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

Each simulated episode is assigned to one stage, consistent with the separate surface and subsurface environments used in training and evaluation. Thus, $m_i(t)$ selects the applicable speed bound and motion law for that episode; the handoff from surface transit to subsurface sampling is represented by the stage initialization, whereas continuous diving, ballast transitions, and resurfacing are not resolved as controlled degrees of freedom.

The horizontal motion of the SUV is modeled as

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{p}_i(t), & m_i(t) = \mathrm{s}, \\ \mathbf{p}_i(t) + \mathbf{u}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m and $\mathbf{u}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Thus, the underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \begin{cases} \mathbf{x}_{i,k} + \mathbf{p}_{i,k}\Delta t, & m_{i,k} = \mathrm{s}, \\ \mathbf{x}_{i,k} + \left[\mathbf{p}_{i,k} + \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)\right]\Delta t, & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

This equation is deliberately a reduced-order horizontal point-mass kinematic–advection model rather than a six-degree-of-freedom rigid-body dynamic model. It does not resolve added mass, hydrodynamic drag, roll, pitch, yaw or heading dynamics, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. This simplification is consistent with the information content of the environmental forcing. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude and actuator states. Introducing a detailed six-degree-of-freedom model would therefore require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. This choice reflects the mission-level scope and available calibration data, not a claim that six-degree-of-freedom effects are operationally negligible. The reduced model retains the first-order effects relevant to mission-level planning: mode-dependent speed limits, propulsion commands, ambient-current advection in subsurface mode, and coastal constraints. Accordingly, the results are interpreted as mission-planning and coordination evidence rather than as predictions of attitude, structural loading, or low-level controller response. Coupling this planning layer to an experimentally calibrated six-degree-of-freedom vehicle model is left for future work.

The local observation of SUV i is

$$\mathbf{o}_i(t) = \left[\mathbf{s}_i(t), \mathbf{v}_i(t), \mathbf{h}_i(t)\right], \tag{4}$$

where $\mathbf{s}_i(t)$ contains the observation used in the corresponding stage and $\mathbf{v}_i(t)$ is the normalized commanded propulsion velocity, $\mathbf{v}_i = \mathbf{p}_i / V_{\max}^{m_i}$; it is not the total ground velocity. In the final surface experiments, $\mathbf{s}_i(t)$ consists only of the normalized wind patches at 10 m above the sea surface sampled by the N SUVs, and no historical frame is used. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater active-control comparison, no inter-vehicle communication is assumed. Accordingly, $\mathbf{s}_i(t)$ consists only of SUV i's local current

speed and components at its assigned ocean depth, while $\mathbf{h}_i(t)$ contains a five-frame history of that same vehicle's local current. Each underwater SUV therefore knows neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. The final surface input also excludes agent-identity encodings and prior-frame history; agents are differentiated only by their current positions, wind measurements, and propulsion velocities. Accordingly, the surface observation dimension is $d_o^{\mathrm{s}} = 3N + 2$. The underwater active controller uses a 1×1 local-current observation plus a five-frame history and two normalized propulsion components, giving $d_o^{\mathrm{u}} = 5 \times 3 + 2 = 17$.

During training, the reward function uses the distance between each SUV and the best-track TC center to shape an association between local 10 m wind patterns and movement toward the center/high-wind region. This is reward-based policy learning, not supervised learning from center labels, and the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The frozen policy receives only the shared fleet-level wind samples and must move toward an unseen TC, thereby testing whether the learned wind-to-inward-motion behavior transfers across events.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{5}$$

and is projected onto the unit disk before being mapped to propulsion in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}\tilde{\mathbf{a}}_i(t). \tag{6}$$

This mapping ensures that the total propulsion speed never exceeds the mode-specific speed limit.

The cooperative task is a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{O}_i\}, \{\mathcal{A}_i\}, P, R, \gamma \rangle, \tag{7}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{O}_i$ and $\mathcal{A}_i$ are the local observation and action spaces of vehicle i, P is the transition probability, r_i is the local reward contribution, $R = (1/N)\sum_{i=1}^{N} r_i$ is the shared team reward, and γ is the discount factor.

3. Data and Simulation Environment

3.1. Reanalysis and Best-Track Products

ERA5 (ECMWF Reanalysis v5) supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [55]. Wind speed is calculated from u_{10} and v_{10}; the policy uses the speed together with the two velocity components. HYCOM (Hybrid Coordinate Ocean Model) supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [56]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS (International Best Track Archive for Climate Stewardship) provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [57]. The common study domain is 98.7°E–171.6°E and 6.8°N–56.7°N. The overall event set spans 19 November 2007 to 3 December 2025 UTC. Thus, the surface observation variable is the 10 m wind velocity and speed, whereas the subsurface observation variables are the horizontal current velocity, represented by `water_u` and `water_v`, and the derived current speed at the seven sampled depths between 0 and 60 m. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the event-specific environments.

3.2. Spatiotemporal Alignment and Track Segmentation

All fields are mapped to a common simulation clock. Surface atmospheric fields are linearly interpolated from hourly values, and ocean currents are linearly interpolated from three-hourly values to the 10 min decision interval. This is an offline reanalysis simulation: adjacent reanalysis samples are used to reconstruct the field at the simulated time, and the policy receives only the interpolated local value at that time, not a future time index or full future field. This construction does not emulate the data latency of a real-time forecast system. At each decision time, the observation operator uses the reconstructed field, while the best-track center is used only for training supervision and evaluation. Any NaN or infinity produced by interpolation or coastal masking is replaced by a pre-specified neutral value after finite-difference gradients are computed. The resulting observations are checked for finiteness before entering the policy.

Although the native grids and sampling intervals are much coarser than the 10 min decision step, the environment performs spatial and temporal interpolation on demand at every decision time. Each SUV queries the interpolated field at its current position and at the corresponding simulation time, and the same interpolated values are used for its observation and for current-induced motion. These values are treated as the best available estimates from the archive under the stated interpolation scheme, rather than as resolved high-resolution truth or statistically optimal reconstructions of unobserved sub-grid variability.

Policy training and evaluation are offline with respect to the environmental data: the ERA5 and HYCOM trajectories are precomputed, and vehicle motion does not feed back into the wind or current fields. This one-way forcing assumption is motivated by the large separation of scales between a small

[1] ERA5 download portal: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels.

[2] HYCOM GLBy0.08 archive: https://www.hycom.org/data/glby0p08/expt-93pt0; HYCOM ESPC-D-V02 archive: https://www.hycom.org/data/espcd-v02.

[3] IBTrACS download portal: https://www.ncei.noaa.gov/products/international-best-track-archive.

Table 1
Data used to construct the TC observation environment. Each event retains its own time window; the common time span is reported for the complete archive. The target archive combines ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

Data product	Source	Variables used	Native grid	Native time
Surface wind at 10 m above sea level	ERA5	u_{10}, v_{10}, derived speed	$0.25° \times 0.25°$	Hourly
Upper-ocean current at 0–60 m depth	HYCOM	`water_u`, `water_v`, speed at 0/10/20/30/40/50/60 m	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Best track	IBTrACS	Position, time, wind, pressure	Point track	Mainly 3/6-hourly

Note: the final subsurface matrix uses paired events for which the selected depth is non-empty. The main policy uses ERA5 winds and HYCOM currents only; temperature and salinity are not policy inputs.

SUV and a TC-scale atmosphere–ocean system, for which the momentum and energy transferred by the vehicles are negligible relative to those driving the environmental fields. It differs from an online two-way coupled simulation in which actions modify the environment and the modified environment then changes subsequent observations. The term offline here does not mean batch or offline RL: the policy is still trained through sequential closed-loop interaction with the precomputed environment, but the environmental trajectory itself is independent of the policy. A two-way coupled simulation would be required if vehicle-induced wakes or other local environmental feedback were the object of study.

The 10 min decision interval is selected as a compromise between control responsiveness and the information content of the source products. A longer interval would reduce the frequency of trajectory correction and degrade decision precision. A substantially shorter interval would increase decision and execution overhead and would substantially increase the number of decisions per episode, the required interaction samples, and the training workload. It would also require temporal detail beyond what can be supported by interpolating hourly ERA5 winds and three-hourly HYCOM currents. Using such an interval would therefore create a false impression of higher temporal resolution without introducing genuinely new environmental information.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111\Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111\Delta\varphi_k\right)^2}, \qquad (8)$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are the longitudinal and latitudinal increments in degrees. The constant 111 km is the mean meridional arc length per degree, and $\cos(\varphi_k)$ converts a longitudinal degree to the corresponding zonal distance at latitude φ_k. The target length of each segment is sampled uniformly from 300–500 km. The start of the next segment is shifted by a fixed 250 km along the track, producing an overlap of 50–250 km and an average overlap of approximately 150 km. A new segment is initialized only when at least 300 km of track remains; the final residual is discarded. Because segment endpoints are assigned to the nearest 10 min interpolation point, the realized segment lengths are 292.1–507.1 km (median 396.4 km).

The 500-TC archive yields 7,248 segments from 495 TCs with sufficient track length. The training split contains 450 TCs, of which 445 contribute at least one usable segment, yielding 6,480 training segments. The held-out test split contains 50 TCs and 768 segments. The training and test sets are separated at the TC level, so segments from the same TC never appear in both sets. Segment durations range from 2.2 to 103 h, with median and mean durations of 16.5 and 20.0 h, respectively.

Track segmentation is not merely a data-processing choice; it reflects the physical operating limit of the surface platform. The surface endurance of an SUV is finite, so one deployment cannot observe the complete life cycle of a TC. A 300–500 km along-track segment instead represents a self-contained mission in which the fleet can correct forecast-induced displacement, reach a physically relevant part of the TC, and collect a sustained observation sequence representative of a bounded surface mission. This segment length is the distance traveled by the TC along its track, not a cap on the total vehicle-route length; turning, lateral correction, and search motion can make an individual SUV path longer. The simulation uses path length as a propulsion cost proxy and does not impose a detailed battery state-of-charge model. After entering the near-core region, the vehicles can, in the mission concept, dive and sample the upper ocean at several depths. The Blue Whale-class platform provides the surface-transit scale and reference depth used in the present model. A future reconfigurable payload could support current profilers, CTD, and biogeochemical sensors, so the observing concept could collect temperature, salinity, current velocity, nutrients, oxygen, and related variables. The control experiments deliberately use current velocity only as the decision-making observation, avoiding model dependence on variables that are not consistently available across the complete 500-TC archive. This restriction does not imply that the physical platform would observe only currents.

4. Multi-Agent Reinforcement Learning Method

4.1. Blue Whale-Class Platform Modeling

The simulated platform state includes horizontal position, propulsion velocity, local surface or subsurface current, accumulated path length, and, when enabled, distance to the coastline. The surface maximum speed is evaluated in

Table 2
Public Blue Whale reference and two simulation scenarios. The two simulation columns are modeling assumptions, not certified vehicle data.

Property	Public reference	High speed scenario	Current speed scenario
Role	Engineering reference	Future SUV	Near-current SUV
Surface speed	23 kn, ≈11.83 m/s	20 m/s	12 m/s
Submerged speed	3 kn, ≈1.54 m/s	1.5 or 2.0 m/s	1.5 or 2.0 m/s
Submerged endurance	≥4 h at 3 kn; ≥72 h standby	20 h scenario	20 h scenario
Sampling depth	≈60 m	0–60 m (7-layer)	0–60 m (7-layer)
Payload	≥800 kg	800 kg	800 kg
Surface fleet size	Single platform	4/8/12/16	4/8/12/16
Decision interval	Not specified	10 min	10 min
Initial surface formation	–	100 km ring, $360°/N$	100 km ring, $360°/N$
Subsurface assignment	Single platform	Four SUVs per depth	Four SUVs per depth

Note: the 20 h entry denotes the simulated observation window under an assumed future endurance and is not a certified 20 h powered-endurance specification for the public reference platform.

a future 20 m/s scenario and a 12 m/s scenario close to the current Blue Whale's public 23 kn. Subsurface maximum propulsion speed is 1.5 or 2.0 m/s. The latter is a clearly marked future-upgrade case. State propagation follows the mode-dependent speed bound and horizontal motion model in Eqs. (1) and (2), with actions mapped to propulsion through Eq. (6). Surface motion neglects the direct current contribution because the 12–20 m/s propulsion speed is much larger than the typical surface-current contribution, and because the surface stage is intended to represent TC approach and high-wind-region tracking. In contrast, the subsurface model explicitly retains ambient advection because the 1.5–2.0 m/s propulsion speed is of the same order as the TC-induced current. The reduced point-mass formulation and its limitations are stated after Eq. (2). Table 2 separates the public engineering reference from the two simulation scenarios and identifies which values are modeling assumptions rather than certified platform specifications.

4.2. Decentralized Partially Observable Decision Process

The cooperative observation task uses the Dec-POMDP defined in Eq. (7). In that formulation, the surface input concatenates the $3N$ normalized wind-observation values acquired across the fleet with the vehicle's own two normalized velocity components; it is memoryless and relies on idealized inter-vehicle communication. In the active underwater setting, each vehicle instead uses only its own current observation, causal five-frame current history, and propulsion velocity. Shared policy parameters therefore do not imply shared observations, states, or actions. In both stages, the actor receives neither the complete environmental field, the global best-track center, nor future reanalysis frames; the best track is used only for training rewards and evaluation metrics.

The policy operates as a closed perception–decision–action loop rather than following a precomputed static route. At every 10 min decision interval, it observes the current environment, generates a new action, moves the vehicle, observes the updated environment, and repeats:

$$\begin{aligned}\mathbf{o}_{i,t} &\longrightarrow \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \\ &\longrightarrow \mathbf{x}_{i,t+1} \longrightarrow \mathbf{o}_{i,t+1}.\end{aligned} \tag{9}$$

This feedback process allows the fleet to revise its motion as the wind and current fields evolve. The observation is not only an output recorded after motion; it is also the next control input that determines subsequent motion. Here, "supervision" refers to this measurement-driven feedback role rather than to supervised learning with target action labels: the policy is trained from scalar rewards, but each new observation directly conditions the next decision through the closed-loop state. Because the environment is nonstationary and stochastic, observations affect subsequent motion continuously rather than being used only to construct a map or waypoint sequence before execution. This distinction applies specifically to planning pipelines that produce and then execute a static route; it does not imply that all model-predictive or adaptive planners are open loop. The same mechanism supports adaptation to changing local observations, partial observability, and deployment uncertainty.

4.3. Surface and Subsurface Reward Design

The surface objective is to follow the translating TC as closely as possible and reach the high-wind region. Its primary operational metric is the final tracking error, defined as the fleet average of the individual SUV-to-center distances at the final step. Let d_i denote the distance from SUV i to the TC center. The surface reward used in the implementation is

$$\begin{aligned}r_i^{\mathrm{surf}} = &-\frac{\min(d_i, 300)}{100} + b_{\mathrm{near}}\mathbf{1}_{d_i<100} \\ &-\frac{W_{\mathrm{form}}\max(0, d_i - R_{\mathrm{form}})}{100} \\ &-\lambda_u\|\mathbf{a}_{i,t} - \mathbf{a}_{i,t-1}\|_2^2 + r_i^{\mathrm{coast}},\end{aligned} \tag{10}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} =$

1. The third term adds an additional radius-excess penalty beyond $R_{\text{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term r_i^{coast} represents the coastal-safety reward defined below. The 300 km cap prevents distant errors from producing an excessively steep gradient and reduces premature policy collapse during early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, four SUVs are assigned to each prescribed depth. Their objective is not to follow the surface TC center directly, but to observe the highest possible current speed. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily the location of the strongest subsurface current. For the active-control comparison, the current-seeking reward is

$$\begin{aligned} r_i^{\text{sub}} = \; & \min\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\text{norm}}}, R_{\max}\right) \\ & - \lambda_E \|\tilde{\mathbf{a}}_i\|_2^2 + \beta_{\|} \max(0, \cos\theta_i) \\ & + r_i^{\text{coast}}, \end{aligned} \tag{11}$$

where $V_{\text{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_E = 0.05$, and $\beta_{\|} = 0.5$. The angle θ_i denotes the direction difference between propulsion and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, $\cos\theta_i = (\tilde{\mathbf{a}}_i \cdot \mathbf{u}_{i,k})/(\|\tilde{\mathbf{a}}_i\|_2 \|\mathbf{u}_{i,k}\|_2)$ when the denominator is nonzero. The current-intensity reward is therefore evaluated at $\mathbf{x}_{i,k}$ before the state update. For the local maximum-current objective, passive drift is retained as the no-propulsion reference because it follows the ambient flow directly and does not depend on a learned reward policy.

4.4. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance D_{buf} receives

$$r_i^{\text{coast}} = \begin{cases} -P_{\text{land}}, & D_i = 0, \\ -W_b \max(0, 1 - D_i/D_{\text{buf}}), & 0 < D_i < D_{\text{buf}}, \\ 0, & D_i \geq D_{\text{buf}}, \end{cases} \tag{12}$$

where D_i is the offshore distance and P_{land} and W_b are positive penalty coefficients.

Three complementary mechanisms are used to reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The current main experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. The deployment step removes invalid initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. These mechanisms reduce land exposure but are not posed as an unconditional hard constraint on every vehicle position.

To reduce oscillatory control, the main surface experiments use an exponential action filter:

$$\mathbf{a}_t^{\text{filt}} = (1 - \beta)\mathbf{a}_t^{\text{raw}} + \beta \mathbf{a}_{t-1}^{\text{filt}}, \tag{13}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions and therefore does not introduce future-information leakage.

4.5. Parameter-Sharing Independent PPO

All SUVs contribute to the same observation and tracking objective, and the training return is defined as the team-average reward. The problem is therefore a fully cooperative task, rather than a competitive or mixed cooperative–competitive task. This cooperative reward structure motivates the use of IPPO, in which decentralized vehicle actions are optimized against a common team objective while policy parameters are shared across vehicles.

In this paper, IPPO denotes a cooperative, parameter-sharing implementation of independent PPO. All SUVs use the same actor and critic weights, while their observations and action samples remain vehicle-specific. Actions are sampled independently, and the per-agent values are summed for training without a global-state centralized critic. The main surface model uses a memoryless multilayer perceptron, whereas the active underwater controller uses a five-frame current history. The shared cooperative return is the team-average reward, and a single clipped PPO objective is optimized over the factorized multi-agent action vector:

$$\begin{aligned} L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(& \rho_t(\theta)\hat{A}_t, \\ & \text{clip}\left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon\right)\hat{A}_t\Big)\Big], \end{aligned} \tag{14}$$

where

$$\rho_t(\theta) = \prod_{i \in \mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\text{old}}}(a_{i,t} \mid o_{i,t})}, \tag{15}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus. During execution, each SUV computes its action from its own observation and the shared policy network.

Off-policy SAC and TD3 were not used for the final multi-SUV comparison. Both require a replay buffer, and the storage per transition increases with the joint observation and action dimensions of up to 16 vehicles. Large-scale parallel sampling also fills replay memory quickly and adds replay-management overhead. The final experiments instead prioritize stable parallel on-policy sampling and use parameter-sharing IPPO. This is an implementation and resource constraint, not a claim that IPPO is universally more sample-efficient than SAC or TD3.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix A. The project repository and software package sources are listed in the Code and package availability section.

4.6. Surface Evaluation and Generalization

The center-distance reward is available only during training. It does not enter the actor observation and is absent during validation. Consequently, the surface results test a specific form of generalization: a frozen policy must infer where to move from shared wind observations in an unseen TC. The policy is not asked to read the best-track center, and it cannot exploit a reward computed from that center. This distinction is essential for interpreting the 100–800 km curves as forecast-error tolerance rather than as supervised center tracking.

Operationally, surface observation benefit is represented by the final tracking error (km), the fleet-averaged SUV-to-best-track-center distance at the final time step. A smaller value denotes a higher benefit because the fleet ends closer to the TC core. Time-averaged distance is retained as a secondary process diagnostic and is computed analogously, with the final-time distance in Eq. (16) replaced by the mean distance over all decision steps of each episode.

For training seed s and its set of $M_s = 768$ held-out segments, the seed-level value is

$$D_{\text{final}}^{(s)} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e}(T_e) - \mathbf{x}_{\text{TC},e}(T_e) \right\|_2 , \quad (16)$$

where T_e is the final decision time of test segment e. The reported final tracking error is the mean of $D_{\text{final}}^{(s)}$ over the ten independent training seeds, with uncertainty expressed as the sample standard deviation across seeds.

The final surface matrix uses only the wind observations collected by the fleet and evaluates 4-, 8-, 12-, and 16-SUV formations. The 100 km ring and equal angular spacing of $360°/N$ define the initial deployment geometry; the formation is not constrained to remain rigid during tracking. During training, the formation-center offset is sampled uniformly inside an 800 km disk around the initial TC center, representing the assumed upper bound of track-forecast position error. The trained policies are then evaluated without retraining at fixed formation-center offsets of 100, 200, 400, 600, and 800 km with random bearings. Because the training distribution already contains radii from 0 to 800 km, the 200 km validation case is covered without a separate 200 km-trained model. Training at only the shortest radius would also remove the deployment variation needed to test transfer across forecast-error levels.

For each fleet-size, deployment-radius, and speed setting, the 768 held-out segments were first averaged within each training seed. Means and sample standard deviations were then computed across the ten independent seeds. Paired comparisons between 12 and 20 m/s used two-sided paired t-tests on the ten seed-level values. Because the twenty fleet-size/radius comparisons were tested separately without multiplicity correction, the resulting p-values are interpreted as exploratory evidence rather than confirmatory hypothesis tests.

4.7. Underwater Evaluation Protocol and Baselines

Once the high-wind region is reached, diving provides the second part of the sampling objective. The subsurface current there is expected to contain the upper-ocean response generated by the TC forcing and its preceding passage, including near-inertial and mesoscale variability [11]. Sampling that dynamically conditioned region is expected to be more informative than descending in an unrelated background ocean. The present study evaluates the observation task itself; it does not quantify the subsequent impact of these measurements in a data-assimilation or forecast system.

The final subsurface matrix evaluates one four-SUV fleet at each ocean depth; the seven depth cases are separate experiments rather than a 28-vehicle fleet. Each episode has a nominal 20 h duration and terminates earlier when the available track window or pre-landfall interval ends. Its objective is to observe the highest attainable local current speed at depths of 0, 10, 20, 30, 40, 50, and 60 m under maximum propulsion speeds of 1.5 and 2.0 m/s. It compares active IPPO, random control, and passive drift under otherwise identical conditions. Because the surface stage is responsible for approaching the TC and high-wind region, the subsurface rollout starts from the surface endpoint and initializes the four vehicles within 200 km of the TC center. The 100–800 km forecast-error protocol applies only to the surface stage.

For each depth and speed setting, one central-index segment is selected from each held-out TC and evaluated under ten paired deployment seeds. The complete 50-TC test split therefore provides 500 matched episodes per policy; the current interim test uses 23 TCs and 230 episodes.

Subsurface execution follows the local-only observation contract defined in Section 2. The policy parameters are shared, but no observations, positions, or actions are exchanged between vehicles during execution. Increasing the fleet at one depth would increase aggregate sampling and change the total observation effort. Because no observations are exchanged, however, additional vehicles would not provide extra information to an individual policy; the resulting comparison would confound strategy with sampling scale. The experiment therefore fixes the fleet at four vehicles per depth and pairs all methods on the same episodes and deployment seeds. In a larger deployment, four vehicles can be assigned to each of several depths, while the marginal value and redundancy of additional vehicles per depth remain a separate design question.

Let $\Pi_{\mathcal{D}}(\mathbf{a}) = \mathbf{a}/\max(1, \|\mathbf{a}\|_2)$ denote projection onto the unit disk. The three strategies differ only in how the propulsion command is generated:

$$\begin{aligned} \mathbf{a}_i^{\text{active}} &= \Pi_{\mathcal{D}}\left(\pi_\theta(\mathbf{o}_i)\right), \\ \mathbf{a}_i^{\text{random}} &= \Pi_{\mathcal{D}}\left(\mathbf{e}_i\right), \qquad \mathbf{e}_i \sim \mathcal{U}\left([-1,1]^2\right), \\ \mathbf{a}_i^{\text{drift}} &= \mathbf{0}. \end{aligned} \quad (17)$$

During evaluation, the active IPPO policy uses its deterministic mean action, whereas random control resamples a two-dimensional action every 10 min. Both commands

are passed through the same unit-disk projection before multiplication by $V^{\mathrm{u}}_{\max}$. Active IPPO and random control share the same propulsion limit and equations of motion; only the active policy conditions its action on the local current history. Passive drift applies zero propulsion and advances solely by ambient current advection.

For vehicle i at depth z, the ground-path length after K decision steps and the mean experienced current speed are

$$L_i(z) = \sum_{k=0}^{K-1} \left\| \mathbf{x}_{i,k+1} - \mathbf{x}_{i,k} \right\|_2 , \tag{18}$$

$$\bar{V}_i(z) = \frac{1}{K+1} \sum_{k=0}^{K} \left\| \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z, t_k) \right\|_2 . \tag{19}$$

$$C(z) = A_{\mathrm{cell}} \left| \bigcup_{i=1}^{N} \bigcup_{k=0}^{K} \mathcal{G}(\mathbf{x}_{i,k}) \right| , \tag{20}$$

where $\mathcal{G}(\mathbf{x})$ is the 0.05° grid cell containing position $\mathbf{x}$, $|\cdot|$ counts unique cells, and A_{cell} is the cell area in km^2. The figures report the fleet means $N^{-1}\sum_{i=1}^{N} L_i(z)$ and $N^{-1}\sum_{i=1}^{N} \bar{V}_i(z)$, while $C(z)$ measures the union of spatial cells sampled by the fleet.

5. Simulation and Results

5.1. Surface Active Tracking

Using the surface protocol in Section 4.6, we evaluate the extent to which the trained policies reduce forecast-induced deployment errors and approach the moving TC center and high-wind region.

Figure 2 reports the primary final tracking error from Eq. (16), for which smaller values are better, and the secondary time-averaged distance. Each value summarizes the 768 held-out segments over ten independent training seeds.

Using the final tracking error as the operational observation benefit, the right panel of Fig. 2 reports the primary speed-limit comparison. Averaged over the four fleet sizes and ten training seeds, the 20 m/s policy obtains final tracking errors of 197.1, 209.2, 260.9, 362.1, and 524.0 km at deployment radii of 100, 200, 400, 600, and 800 km, respectively. The corresponding 12 m/s values are 221.0, 238.9, 317.5, 450.9, and 635.1 km. Thus, 20 m/s yields a lower final tracking error in all 20 matched fleet-size/radius comparisons, corresponding to a higher operational observation benefit. Its fleet-averaged advantage increases from 23.8 km at 100 km to 111.1 km at 800 km.

The absolute distances still increase with deployment radius for both speed limits, which is expected because a finite propulsion speed cannot eliminate a larger initial displacement within every segment duration. The objective is therefore to minimize the final tracking error at each error level rather than to make it independent of deployment error. The final-tracking-error advantage of 20 m/s is larger than the corresponding time-averaged advantage: 23.8 versus 10.1 km at 100 km and 111.1 versus 84.2 km at 800 km. This indicates that the higher-speed policy continues correcting residual separation late in the episode rather than merely remaining closer during the initial transit.

This benefit involves a clear propulsion trade-off. Relative to 20 m/s, the 12 m/s policy reduces path length per SUV by 78.7, 95.5, 147.4, 196.6, and 233.3 km at the five radii and lowers mean ground speed by 1.74–4.18 m/s. Accordingly, 20 m/s provides the higher final observation benefit, whereas 12 m/s is preferable when shorter trajectories and lower propulsion use are prioritized.

Time-averaged distance provides a secondary process-level diagnostic of whether proximity is maintained throughout an episode. Across the twenty matched fleet-size/radius comparisons, the 20 m/s policy has the smaller mean time-averaged distance in all 20 cases. The median paired difference ($d_{12} - d_{20}$) is +51.8 km. Uncorrected two-sided paired tests indicate lower distance at 20 m/s in 19 cases and at 12 m/s in none of the cases at $p < 0.05$. Because these comparisons were not adjusted for multiplicity, they provide exploratory rather than confirmatory evidence.

The relative time-averaged distance penalty of the 12 m/s policy is +6.3% at 100 km, +10.6% at 200 km, +18.2% at 400 km, +17.7% at 600 km, and +14.0% at 800 km. The non-monotonic relative pattern, together with the monotonic increase in absolute final-tracking-error benefit, is consistent with the finite time available to remove forecast-induced displacement. Overall, the results show that the policy learned from wind observations alone provides increasing operational value for large initial displacement errors when a 20 m/s speed envelope is available, while the 12 m/s envelope offers a lower-effort alternative.

Taken together, Figs. 2–4 and Table 3 connect the operational benefit metric to both population-level statistics and representative trajectories. The right panel of Fig. 2 is the primary benefit comparison: for each deployment radius, the preferred configuration is the one with the smallest final tracking error. Figure 3 shows the mechanism in the four-SUV, 400 km case, where the formation first turns along the observed wind structure and then reduces its separation from the TC center. Figure 4 shows that this inward correction persists from 100 to 800 km, although the final marker remains farther from the center as the initial deployment error increases. These representative paths explain why an effective controller can reduce error without making the final tracking error independent of deployment radius.

Table 3 isolates the execution-layer filter on the same trained actors. This separates changes caused by trajectory smoothing from changes caused by policy retraining and allows tracking error to be assessed together with path length and speed.

To provide an internally consistent visual comparison, Figs. 3–5 use the same held-out CHAMPI (2015) segment. The surface figures show motion from the segment start to its endpoint, while the subsurface rollout starts from the

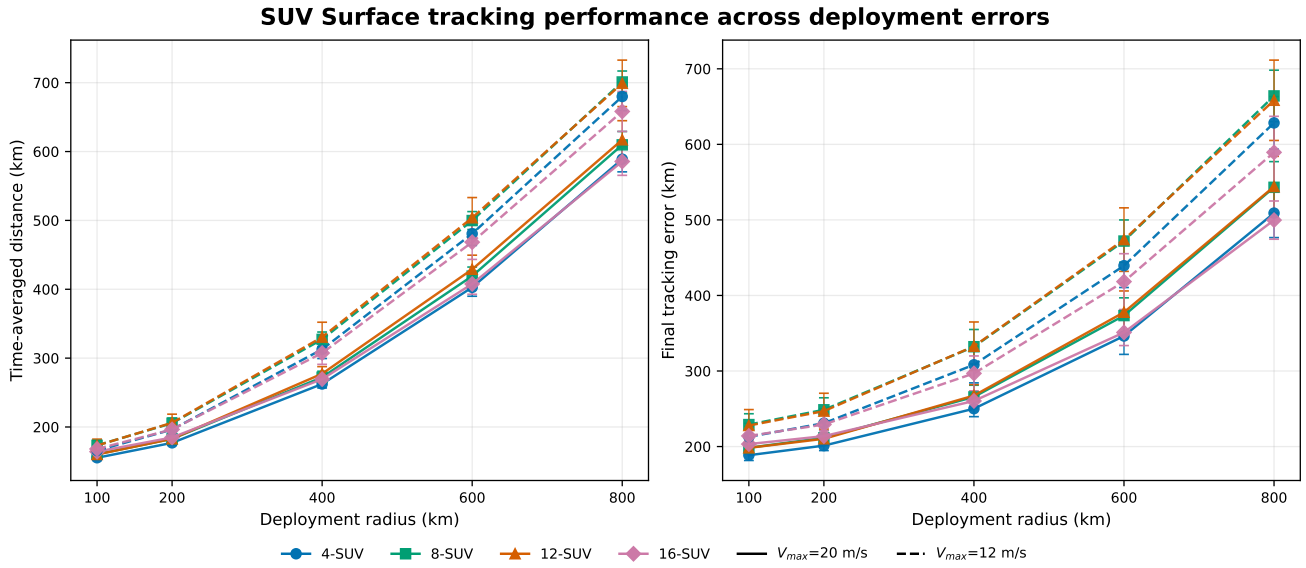


Figure 2: SUV surface tracking performance across deployment errors. The left panel shows the fleet-mean time-averaged distance to the TC center, representing whether the fleet remains close to the moving TC throughout an episode. The right panel shows the final tracking error to the TC center at the end of the episode; smaller values indicate higher operational observation benefit. Colors denote fleet sizes and line styles denote the 20 and 12 m/s speed limits. Each point summarizes ten independent training runs (ten random seeds), with error bars showing the sample standard deviation.

segment-end anchor, representing the sequential transition from surface delivery to underwater observation.

Figure 3 resolves the temporal evolution for the four-SUV, 400 km case. At deployment, the four vehicles are dispersed around the formation center far from the TC. During the first several hours, the vehicles turn toward the strengthening wind region and begin to reduce the initial displacement. By 6.5 h, all four vehicles have initiated the inward turn; by 13.0 h they have entered the outer high-wind region; and by 19.5 h all four trajectories have turned into the high-wind branch and substantially reduced their initial displacement. The individual paths differ because each vehicle has its own state and position within the shared fleet-level wind observation; nevertheless, the fleet-level motion follows the wind pattern toward the TC core and illustrates long-range error correction in this representative episode.

Figure 4 shows how initial forecast error changes the surface search process. At 100–200 km, the formations converge rapidly and remain concentrated around the TC core. As the deployment radius increases to 600–800 km, the trajectories lengthen and fan out over a wider region before turning toward the TC. Increasing the fleet from 4 to 16 SUVs produces a denser set of sampled branches and broader spatial coverage, but it does not remove the need for long-range transit. In the 12- and 16-SUV cases, many paths overlap near the core because the shared wind observations guide the vehicles toward the same high-wind region. In these representative cases, fleet size mainly broadens the sampled approach region, whereas deployment error primarily determines the required transit distance.

The 600 and 800 km rows are also consistent with two coupled limitations at large initial displacement. At 600 km, the representative fleet can still recover a usable inward direction from the observed wind-speed field and its spatial gradient, so the vehicles continue toward the TC despite the longer transit. At 800 km, the vehicles must first overcome a larger geometric separation with finite speed and a finite episode duration; in addition, the TC-induced wind speed and its local gradient become weaker and less distinctive at that range. The reduced signal quality makes the approximate bearing toward the TC more ambiguous, so the policy may approach more slowly, change direction repeatedly, or fail to recover the TC within the episode. These two effects can reinforce each other: longer travel time increases the consequence of even a small directional error, while the weaker remote wind signature makes such an error more likely.

5.2. Execution-Layer Action Filtering

Table 3 reports the low-pass filtering ablation over the same trained actor, 768 held-out segments, and 10 min decision interval. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1294.9–1331.5 km per vehicle, with mean speeds of 19.23–19.73 m/s and maximum speeds of 19.92–20.00 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 690.8–769.8 km, mean speed to 11.11–12.72 m/s, and maximum speed to 16.13–18.20 m/s. The final tracking error changes from 288.2 to 258.2 km for four SUVs, from 305.2 to 308.6 km for eight SUVs, from 478.5 to 330.0 km for 12 SUVs, and from 313.6 to 273.4 km for 16 SUVs. Averaged over the four fleet sizes, it decreases from 346.4 to 292.6 km, a reduction of 53.8 km (15.5%). Thus, filtering not only avoids degrading final tracking but

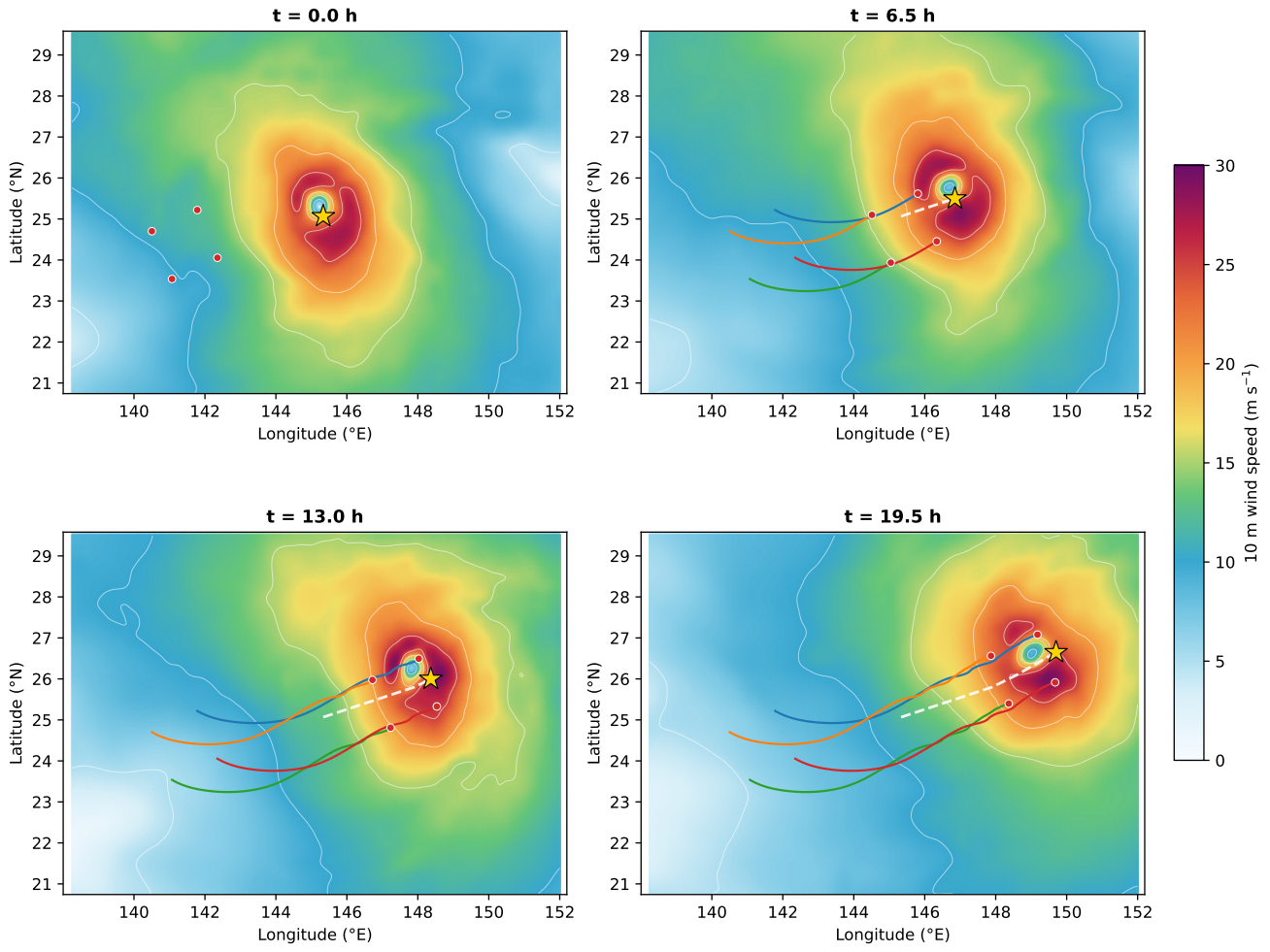


Figure 3: Four-SUV surface-tracking process for CHAMPI (2015) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. Small red dots mark vehicle positions, and the yellow star marks the TC center.

improves it on average: three fleet sizes improve, whereas the eight-SUV case increases only slightly, by 3.4 km.

The same comparison shows reductions for every fleet size: path length decreases by 563.4, 547.1, 618.5, and 573.6 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 1.80, 2.29, 3.87, and 2.94 m/s. The 12-SUV configuration has the largest relative path-length reduction, although the effect is not monotonic in fleet size. These results show that the execution filter improves final tracking on average while substantially shortening trajectories and reducing mean and peak speed.

5.3. Speed-Limit and Deployment-Error Analysis

The 12 and 20 m/s cases answer different engineering questions. The 12 m/s value is close to the published surface-speed scale of the current Blue Whale, whereas 20 m/s represents a future high-speed scenario. A simple direct-pursuit estimate illustrates why deployment radius changes the effect of this choice. If the TC translates at speed U_c and the SUV moves toward it at V_{max}, the nominal closure rate is approximately

$$V_{close} = V_{max} - U_c. \tag{21}$$

For a deployment error D, the corresponding correction time is

$$t_{corr} \approx \frac{D}{V_{close}}. \tag{22}$$

This is only a kinematic lower bound because it neglects turning and imperfect center estimation, but it exposes the dominant scale. At 100–200 km, both speed limits provide enough closure to correct a moderate forecast displacement. At 600–800 km, a TC translation speed of 4–7 m/s removes a substantial fraction of the 12 m/s speed budget. The required correction time can then approach or exceed the 20 h mean segment duration. A 20 m/s vehicle retains a larger reserve for reducing initial center error, but its benefit is expected

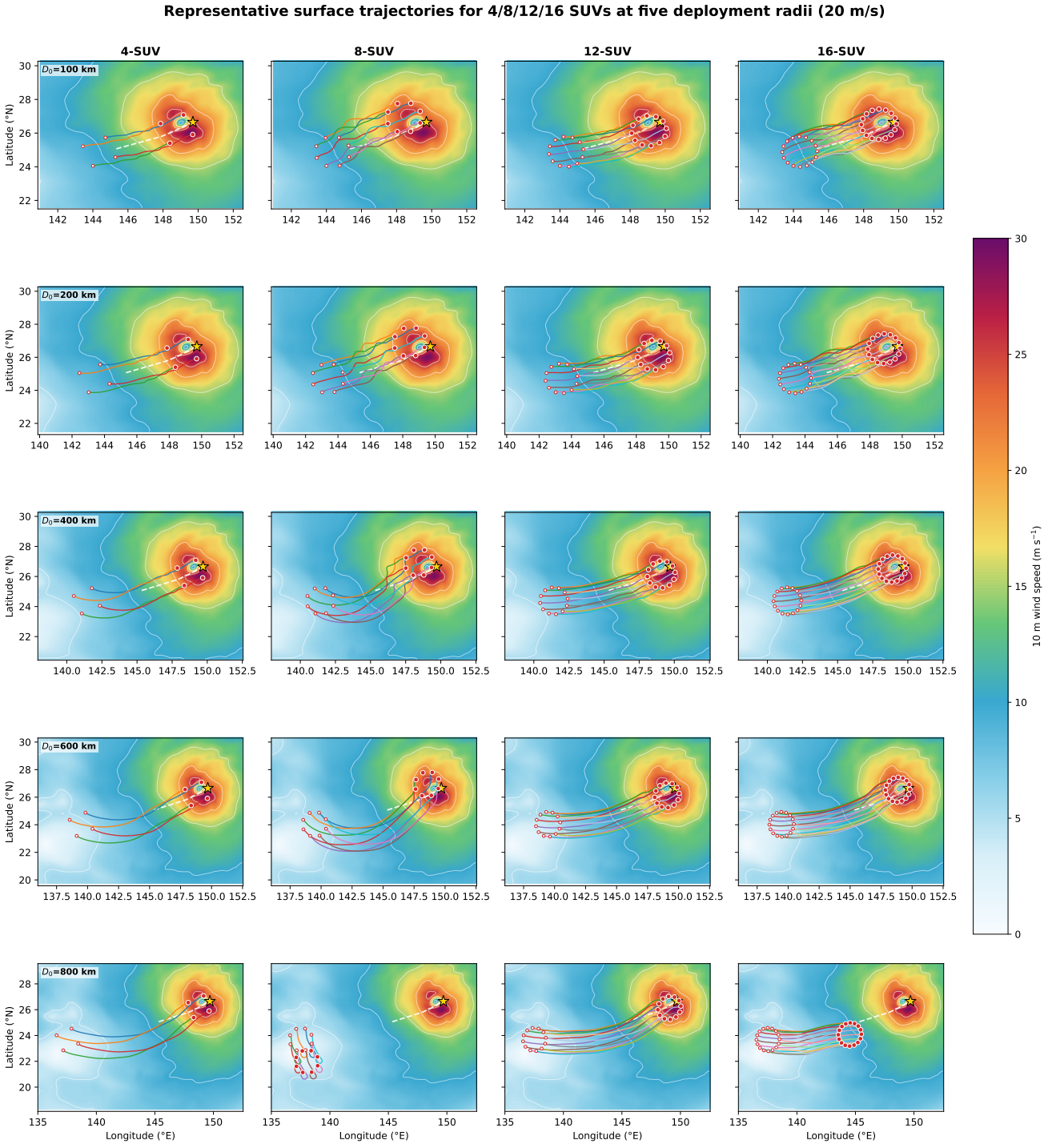


Figure 4: Representative CHAMPI (2015) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show nominal formation-center deployment radii of 100, 200, 400, 600, and 800 km. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC center at the end of the episode.

mainly at the larger radii; at short radii both cases already have sufficient speed.

The growing 12/20 m/s final-tracking-error gap follows this closure-rate scaling: both speeds are adequate at 100–200 km, whereas the available reserve of the 12 m/s platform contracts at 400–800 km. The 800 km test is therefore conservative for the lower-speed platform. Operationally, large forecast errors favor a faster vehicle, a tighter forecast-error envelope, or earlier deployment; 12 m/s remains preferable only when shorter paths and lower propulsion use are prioritized.

Table 3
Effect of execution-layer low-pass filtering for 4/8/12/16 SUVs on 768 held-out segments from 50 unseen TCs in the 500-TC archive. The same trained policy is evaluated with the filter off and on ($\beta = 0.8$).

Fleet	Filter	Final tracking error (km)	Path per SUV (km)	Mean speed (m/s)	Maximum speed (m/s)
4	Off	288.2	1331.5	19.73	20.00
4	On	**258.2**	**768.1**	**12.72**	**18.20**
8	Off	**305.2**	1318.0	19.57	19.97
8	On	308.6	**770.9**	**12.58**	**17.68**
12	Off	478.5	1309.3	19.34	20.00
12	On	**330.0**	**690.8**	**11.11**	**16.13**
16	Off	313.6	1295.0	19.23	19.92
16	On	**273.4**	**721.4**	**11.94**	**16.98**

Final tracking error is the fleet average of the individual SUV-to-best-track-center distances at the final time step. Off/on changes only the execution-layer filter on the same trained actor.

5.4. Underwater Strategy Comparison

For each depth, four SUVs seek higher current speed; the central comparison is therefore the fleet-mean current speed experienced along each trajectory, together with its path and propulsion cost. The comparison follows the protocol in Section 4.7. Figure 5 first illustrates the resulting trajectory structure for the three strategies.

Figure 5 illustrates the qualitative trajectory construction for the three strategies at shallow, intermediate, and deep sampling layers. Across these three representative depths, the mean experienced current speeds of active control, random control, and passive drift remain close within each depth, whereas their path lengths differ substantially: active control requires the longest paths and passive drift the shortest. The background fields also show that the spatial current structure changes with depth, which is why a single-depth example is insufficient. These episodes remain illustrative; all comparative inference is based on the ensemble results in Fig. 6 rather than on these individual trajectories.

Figure 6 reports an interim seven-depth comparison. To keep the depth comparison internally consistent, all policies are evaluated on the same 23 held-out TCs. The 40 and 60 m active policies were trained with the currently available archives (200 and 203 training TCs), whereas the remaining depths use the complete training partition. The plotted values summarize ten independently trained active policies per depth and speed setting. At all seven depths and both propulsion limits, active IPPO attains a higher mean experienced current speed than passive drift and random control. The improvement over passive drift ranges from 1.6% to 5.6%, indicating that the learned policy spends more of each episode in relatively high-current water. Random control remains close to passive drift in mean current speed but travels substantially farther.

The depth dependence of the current metric is consistent across all three strategies. From 0 to 60 m, the mean experienced current speed decreases monotonically from 0.526 to 0.273 m/s for the 2.0 m/s active policy (48.1%), from 0.514 to 0.276 m/s for the 1.5 m/s active policy (46.3%), and from 0.504 to 0.269 m/s for passive drift (46.6%). The similar absolute decay under trained, random, and unpropelled motion indicates that this trend is primarily an environmental signature rather than a policy artifact. It is consistent with the upper-ocean response to TC forcing: wind stress transfers momentum most strongly in the surface mixed layer, after which vertical momentum flux, shear instability, turbulent dissipation, and mixing reduce the horizontal current amplitude with depth [11]. The finite currents at 50–60 m also show that the response persists throughout the sampled layer rather than vanishing abruptly.

The longer trajectories of active IPPO represent a sampling trade-off rather than an unconditional gain. Because all strategies sample at the same 10 min interval, path length does not by itself increase the number of measurements per SUV. It does, however, disperse those measurements over a larger spatial footprint and reduce repeated sampling of nearly the same water mass. The active path is 3.7–7.8 times that of passive drift, and the corresponding coverage increases by roughly 3.3–6.2 times. Active control must therefore be assessed against both objectives: a modest increase in mean experienced current speed and broader spatial context, offset by greater propulsion use. Passive drift remains the low-motion reference and cannot deliberately relocate toward higher-current regions.

Path length follows the same environmental attenuation only weakly. Passive drift has no propulsion term, so its path contracts by 50.8% from 0 to 60 m, almost matching the 46.6% decrease in its experienced current. Active paths decrease by only 17.7% and 19.3% at 2.0 and 1.5 m/s, respectively, because the commanded propulsion component is independent of depth and dominates the 2.0 m/s case. Consequently, the active-to-drift path ratio grows from 4.69 to 7.84 at 2.0 m/s and from 3.69 to 6.06 at 1.5 m/s. This increase should not be interpreted as the policy becoming progressively stronger with depth: it also reflects the faster shrinkage of the passive-drift denominator. The small absolute current advantage remains positive at every depth (1.6–5.6%), while the absolute current itself declines. Thus, upper-ocean current attenuation sets the background depth trend, and active propulsion mainly preserves mobility and sampling extent as that background weakens.

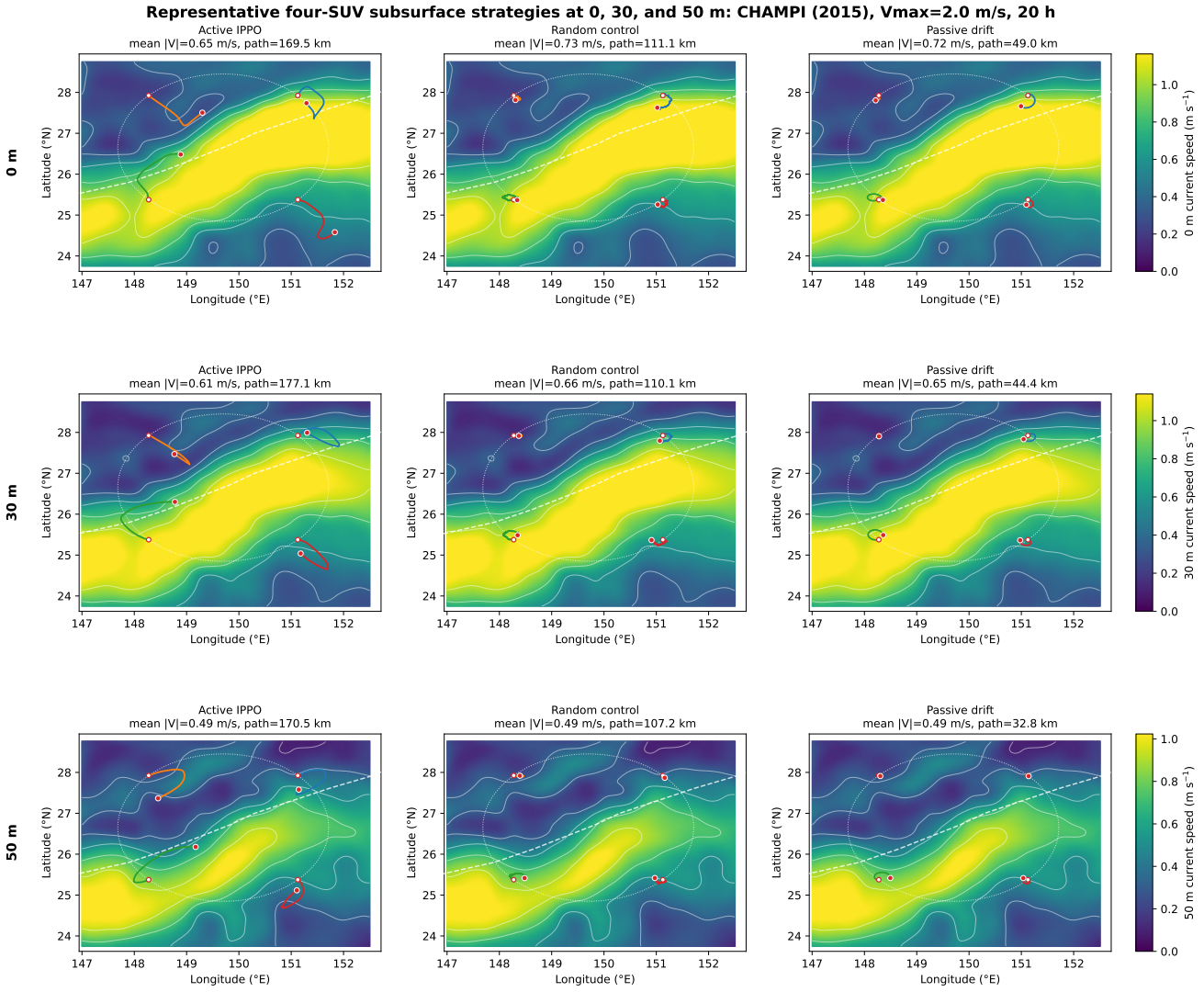


Figure 5: Representative four-SUV trajectories at 0, 30, and 50 m for the same CHAMPI (2015) segment used in the surface trajectory figures. These three depths provide shallow, intermediate, and deep examples from the full 0–60 m evaluation matrix. Rows denote depth and columns denote Active IPPO, random control, and passive drift. The propulsion limit is 2.0 m/s and the nominal episode duration is 20 h. Background color is the spatially interpolated episode-mean HYCOM current speed at the corresponding depth; dashed white curves show the TC best track, open circles mark the common 200 km initial ring, and small red dots mark final vehicle positions.

The propulsion limit consistently affects trajectory extent. At all seven depths, the 2.0 m/s policy travels 1.24–1.32 times farther than the 1.5 m/s policy because the larger speed envelope permits more movement within the same nominal 20 h mission. This result links future platform speed to potential sampling capacity: a faster SUV can cover more of the TC-induced current field and place successive samples farther apart, although the associated energy and control constraints must still be considered. The mean experienced current is slightly lower at 2.0 m/s than at 1.5 m/s at 30 and 60 m, however. A larger speed envelope therefore enlarges the sampling footprint but does not guarantee a higher mean current exposure.

6. Conclusion

This study presented a parameter-sharing IPPO framework for coordinated surface and subsurface TC observation with Blue Whale-class SUVs. Event-specific environments aligned ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks for a 500-TC archive. Training and testing were separated at the TC level to estimate transfer to unseen events.

At the surface, the 20 m/s policy produced lower final tracking error than 12 m/s in all 20 fleet-size/radius comparisons, and its advantage increased with deployment error. The 12 m/s policy instead yielded shorter, lower-speed trajectories. Low-pass filtering further reduced fleet-mean final tracking error and peak speed, showing that wind-only

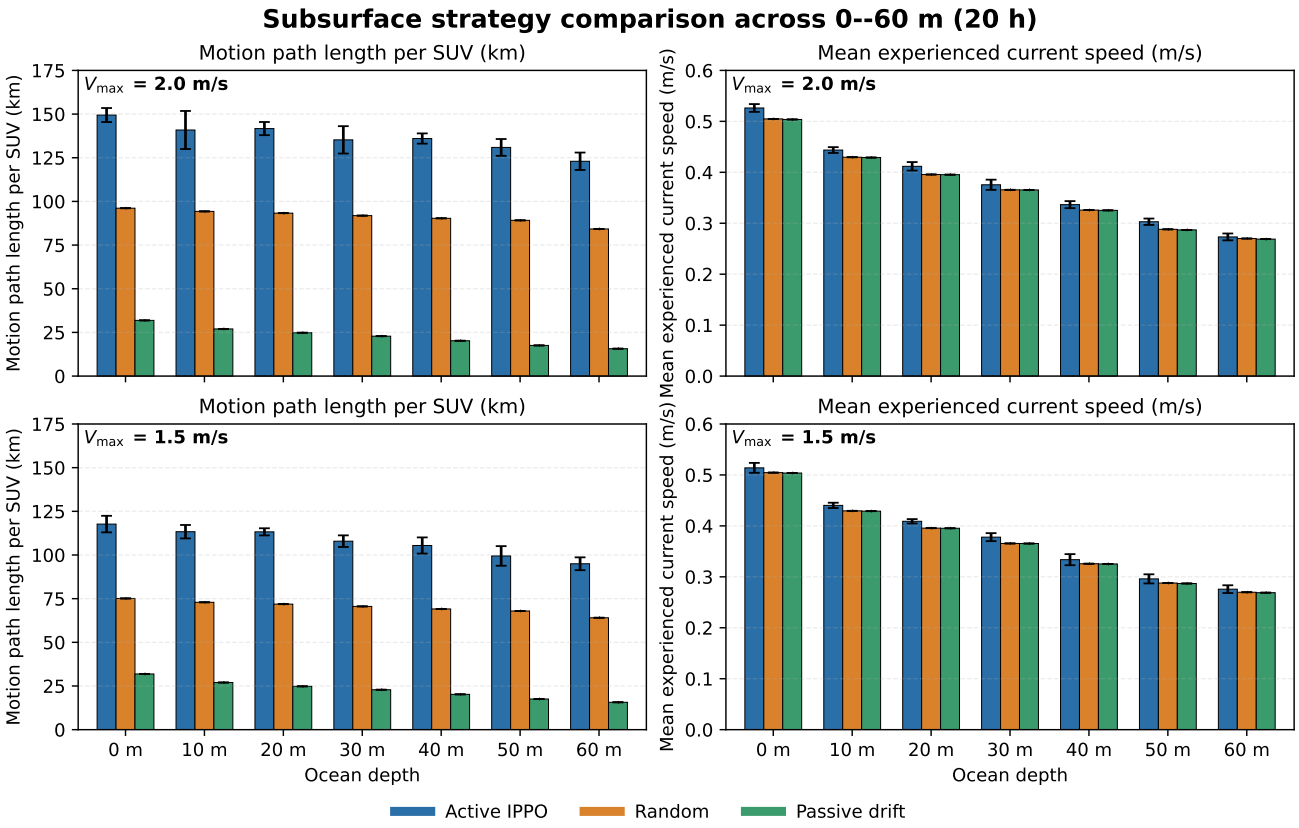


Figure 6: Interim subsurface strategy comparison across seven depths at maximum propulsion speeds of 2.0 and 1.5 m/s. All methods use the same 23 held-out TCs. The 40 and 60 m policies use the currently available archives, containing 200 and 203 training TCs, respectively; the other depths use the complete training partition. Rows show maximum propulsion speed, and columns show the fleet-mean motion path length and mean experienced current speed. Error bars show the sample standard deviation across ten active-policy training seeds. Random control and passive drift are paired baselines without training-seed variability.

observations support closed-loop surface correction without direct access to the TC center.

In the interim seven-depth comparison, active IPPO increased mean experienced current speed by 1.6–5.6%, path length by 3.7–7.8 times, and sampled area by 3.3–6.2 times relative to passive drift. The longer paths represent broader, less redundant spatial sampling at the cost of greater propulsion. Current speed and drift displacement decreased with depth, consistent with the vertical attenuation of TC-generated upper-ocean currents, whereas active propulsion maintained mobility as the ambient flow weakened. At 2.0 m/s, active paths were 1.24–1.32 times longer than at 1.5 m/s, indicating that future speed upgrades could expand subsurface sampling capacity subject to energy and control limits.

The present evidence remains simulation-based and uses a mission-level horizontal point-mass model. The 40 and 60 m results are interim until their training archives are complete; field validation, operational forecast impact, and calibrated six-degree-of-freedom vehicle dynamics remain future work.

Code and package availability

The paper-facing code, environments, evaluation scripts, and processed results are available at https://github.com/DYHAI/SUV-MARL-TC. Experiments used Python 3.11, Stable-Baselines3 2.9.0, Gymnasium 1.2.2, PyTorch 2.11.0+cu128, NumPy 2.4.6, pandas 2.3.3, xarray 2026.7.0, SciPy 1.17.1, and Matplotlib 3.11.1 on a 24-core workstation with 64 GB DDR5 memory and an NVIDIA RTX 5060 Ti 16 GB GPU.

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A. Reward and IPPO Hyperparameters

Table A.1 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table A.2 lists the shared IPPO network and training hyperparameters.

Table A.1: Reward, action-filter, and safety parameters used in the final surface and underwater experiments.

Parameter	Symbol	Value	Description and applicable task
Panel A: surface reward and controller parameters			
Proximity-cost distance cap	$D_{\max}$	300 km	Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments.
Near-center threshold	d_{near}	100 km	Threshold defining the discrete near-center bonus.
Near-center bonus	b_{near}	1.0	Bonus assigned when an SUV is within 100 km of the TC center.
Radius-excess threshold	R_{form}	100 km	Reference radius used by the additional distance penalty beyond 100 km.
Radius-excess weight	W_{form}	1.0	Weight of the penalty applied when an individual SUV is farther than 100 km from the TC center.
Action-change weight	λ_u	0.05	Weight of the squared action change $\|\mathbf{a}_{i,t} - \mathbf{a}_{i,t-1}\|_2^2$ at the 10 min interval.
Action-filter coefficient	β	0.8	Coefficient of $\mathbf{a}_t^{\text{filt}} = (1-\beta)\mathbf{a}_t^{\text{raw}} + \beta\mathbf{a}_{t-1}^{\text{filt}}$. This is a controller setting rather than a reward term.
Surface reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 2]$	Clipping range applied to the team-averaged surface reward.
Panel B: underwater active-control comparison parameters			
Current-speed normalization	V_{norm}	1.0 m/s	Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\max})$.
Current-reward cap	$R_{\max}$	2.0	Upper bound of the normalized current-speed reward.
Propulsion-energy weight	λ_E	0.05	Weight of the propulsion penalty $\lambda_E\|\tilde{\mathbf{a}}_i\|_2^2$ in the current-seeking reward.
Downstream-alignment weight	$\beta_{\parallel}$	0.5	Weight of the positive propulsion–current alignment term.
Underwater reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 5]$	Clipping range applied to the team-averaged underwater reward during active control training.
Panel C: coastal-safety parameters			
Land penalty	P_{land}	20	Penalty applied when an SUV enters a land cell. Shared by the surface and underwater environments.
Nearshore buffer distance	D_{buf}	100 km	Width of the soft nearshore safety buffer.
Nearshore penalty weight	W_b	8.0	Maximum soft-penalty weight within the 100 km nearshore buffer.
Minimum deployment offshore distance	D_{deploy}	100 km	Minimum coastline distance required by surface deployment rejection sampling.
Action-filter safety margin	D_{filter}	100 km	Surface-only distance below which the onshore component of the commanded action is removed.

Note: the final subsurface comparison consists of active IPPO, random control, and passive drift. Passive drift does not optimize a learned propulsion policy or reward.

Table A.2: Parameter-sharing IPPO network and training hyperparameters used in the final surface and subsurface comparisons.

Parameter group	Value	Description
Algorithm	Parameter-sharing PPO (IPPO)	Stable-Baselines3 PPO with a shared actor–critic, independent Gaussian sampling for each SUV, and team-average reward.
Networks	2×256 MLP	Actor and critic each use two 256-unit Tanh layers; per-agent critic values are summed for training.
Surface input	$d_{\mathrm{obs}} = 14/26/38/50$	Wind-only inputs for 4/8/12/16 SUVs: all fleet wind values plus the vehicle's own normalized propulsion velocity; no history or identity input.
Underwater input	$d_{\mathrm{obs}} = 17$	Own current speed and components over five causal frames plus two normalized propulsion components; no inter-vehicle communication.
Action	2 per SUV	Continuous east–north command in $[-1, 1]^2$, projected onto the unit disk before applying the mode-specific speed limit.
Vectorized training	$N_{\mathrm{env}} = 4$	Four parallel environments per training run.
PPO rollout update	2048 / 256 / 10	Rollout length / mini-batch size / optimization epochs.
Optimization	Adam, $\alpha = 3 \times 10^{-4}$	Learning rate for the actor and critic.
Discounting and GAE	$\gamma = 0.99$, $\lambda_{\mathrm{GAE}} = 0.95$	Applied to the shared team-average reward.
Clipping	$\epsilon = \epsilon_v = 0.2$	Policy-ratio and value-function clipping ranges.
Entropy, value, gradient	$c_{\mathrm{ent}} = 0.01$, $c_v = 0.5$, $g_{\max} = 0.5$	Entropy coefficient, value-loss coefficient, and global gradient-norm limit.
Run length and seeds	2×10^6, seeds 0–9	Nominal environment steps and ten independent seeds per trained configuration.
Surface training distribution	$0 \le r \le 800$ km	Formation-center offset is sampled uniformly inside the disk around the initial TC center; this radius is the assumed upper bound of track-forecast error.
Surface validation	100/200/400/600/800 km	Fixed formation-center offsets with random bearings, evaluated without retraining.
Surface geometry	4/8/12/16; 100 km ring	Fleet sizes and initial $360°/N$ ring spacing, updated every 10 min.
Underwater matrix	4 SUVs; depths 0–60 m	Seven independent depth cases at 0/10/20/30/40/50/60 m, each with Active IPPO, random control, and passive drift.
Underwater initialization	$r_0 \le 200$ km	Vehicles start uniformly within 200 km of the TC center; the campaign lasts at most 20 h and may terminate at the data or pre-landfall boundary.
Speed limits	Surface 12/20 m/s; submerged 1.5/2.0 m/s	Two surface and two submerged scenarios after unit-disk action projection.
Compute	CPU	PPO/IPPO policy training and evaluation.

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